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Contents.

Covering everything from the basics, to all things complex.

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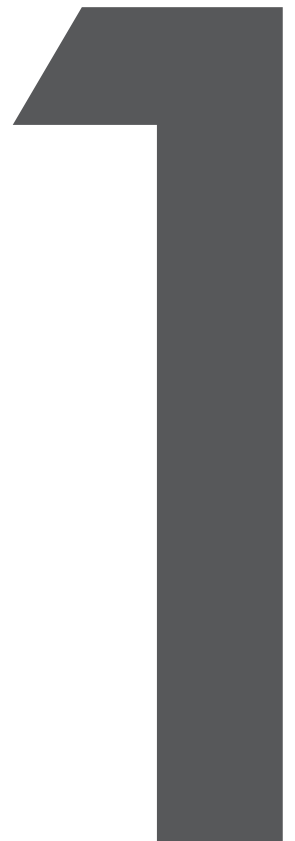
Understanding Typeface Design.

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Typeface Design.

A typeface “is the collection of thousands of years of collective human social, technological and economic history, combined with the passion, skill, experience and personal history of its lone creator.” (Earls, 2002)



An Introduction to Typeface Design

If you're a fan of type and lettering and feel up for a considerable yet potentially rewarding design challenge, then you might consider designing a typeface. Then, should you want a further challenge, you might also consider ways of digitising your work. There are a number of ways that you can approach this—some being far easier than others. Whatever approach you decide to take, a certain amount of patience and understanding will go a long way.

As an art form and a technical, engineering process, typeface design has, in recent years, grown exponentially. A process that has traditionally been the esoteric knowledge of the few, it continues to grow and diversify. Opinions regarding the expanding growth of typeface design have been met with conflicting views. The late English lettering artist and designer, Michael Harvey, expressed his dislike for the “confusing variety of styles” that add to the “overcrowded repertoire” of new typeface designs and the flourishing eccentricity and ‘crumbling’ standards of design. However, the argument about the choice or surplus of available fonts (depending upon your stance), is perhaps better summed up by Peter Bil’ak in his article “We don’t need new fonts...”.

In his piece, Bil’ak states:

“There are typefaces which haven’t been made yet and which we need. Type that reacts to our present reality rather than being constrained by past conventions.... It is time to think about why we design type, not just how we design it.”

Getting Started

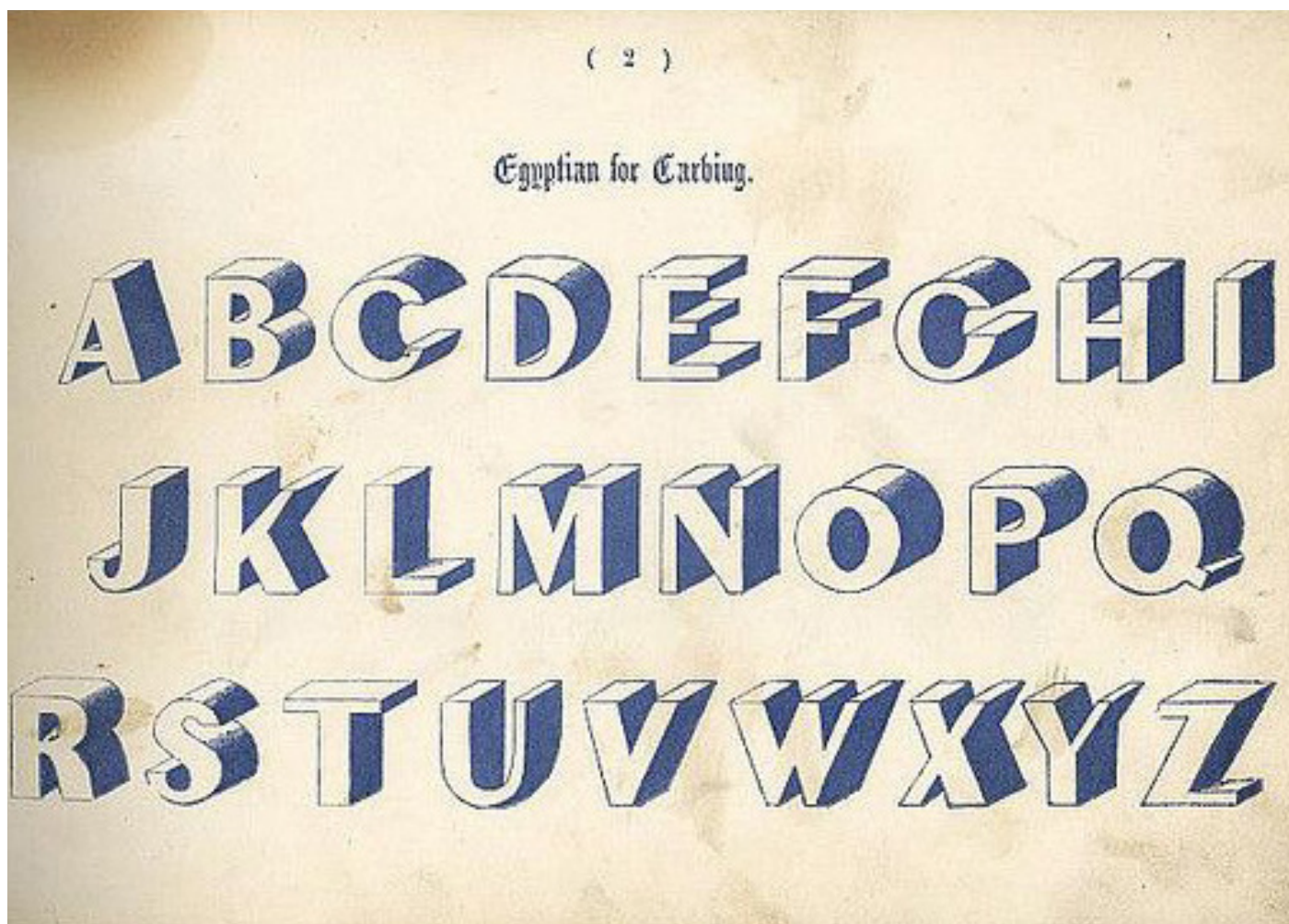
When you begin designing a new typeface, the Internet, along with old or out of print books, can provide a good starting point for your research. Don’t feel obliged to sit at your desk during this process. Get yourself outside and look out for sources of potential inspiration and keep a visual record of the items that inspire you. Importantly, consider the function of the fonts that interest you. What is it that you think the designer set out to do with the type? Is it purely decorative or does it serve a particular function, such as assisting with way finding or traffic signs? With that in mind, consider what you want it to do. You should keep in mind the maxim, *“form follows function”*.

Once you’ve established what the typeface is for, you can begin. If you’re new to the process, you’ll want to keep things simple. Typeface design can be a complex enough task without making it more difficult than it needs to be. Perhaps you should start with a simple display face. Depending upon the requirements of a brief, some designers spend months (even years!) perfecting their work, so it’s a task that can easily grow in scale.

Some designers begin working by drawing letters, keeping sketchbooks and transferring these designs to the computer. Other designers work directly within a digital environment. For the latter, FontStruct is an ideal place to begin, being an easy to use, free, online resource, but we will address technological resources at a later point.

On finding inspiration, the French typeface designer Jean-Francois Porchez suggests we must stand on the shoulders of giants for inspiration and knowledge, stating:

“Each new typeface cannot live without its predecessors – all of them form one large family and are connected and influenced by each other. If Claude Garamond cut a splendid type, then it’s because he studied Italian typefaces and tried his best to improve upon them...”



A B C D E

F G H I J K

L M N O P

Q R S T U

V X Y & W Z
W \$_{del}

Terminology & Anatomy.

"There are typefaces which haven't been made yet and which we need. Type that reacts to our present reality rather than being constrained by past conventions.... It is time to think about why we design type, not just how we design it." (Peter Bil'ak)



Even if you are an experienced calligrapher or lettering artist, making a typeface is an altogether different process, and before you start it's worth becoming familiar with some fundamental typographic concepts and terminology.

You have probably heard of the terms fonts and typefaces, and these two terms are often used interchangeably as there seems little harm in doing so. However, strictly speaking, there is a distinct difference in meaning between the two.

A typeface is essentially the design of an alphabet and its family, such as Gill Sans Regular, Bold and Italic in its various sizes (12 point, for example), whereas a font or “fount” (UK) is the physical collection of a typeface and the means by which it is rendered. For example, either the digital or metal version of Gill Sans Bold 14 point is a font.

Another simple example of these differences is stencil lettering and the means by which it is produced. The design of the letter shapes on the stencil is the typeface and the plastic stencil used to reproduce the design is essentially the font.

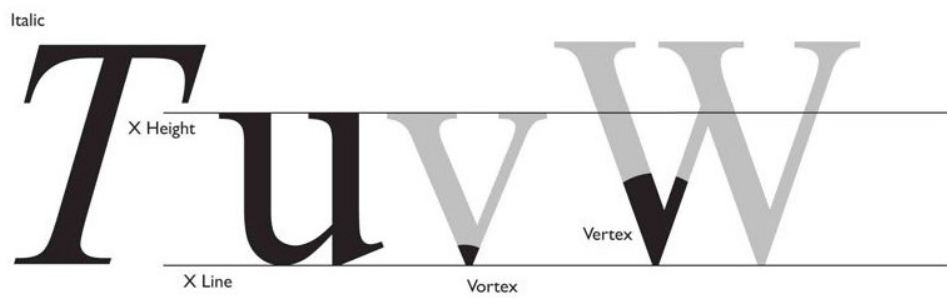
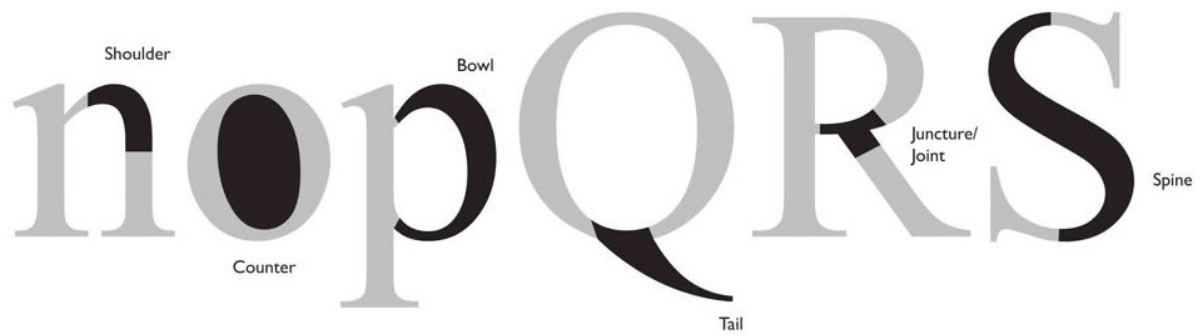
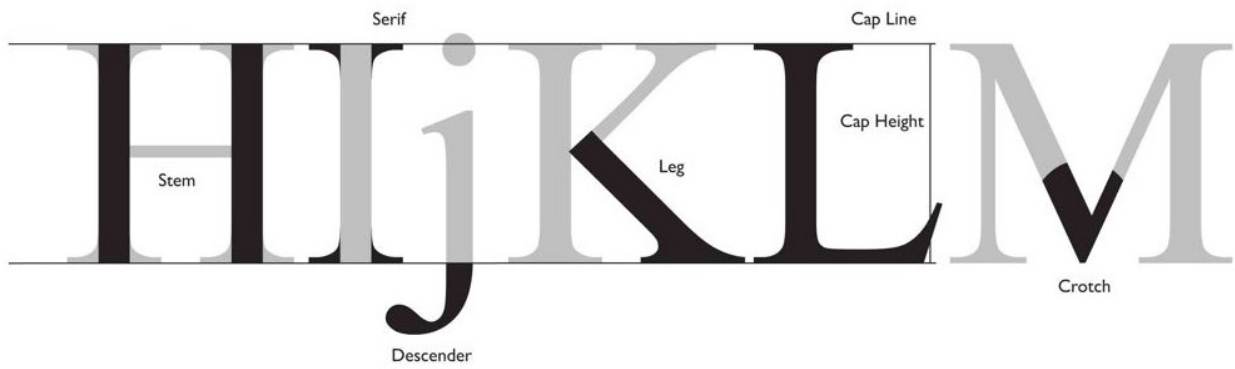
Where to Start?

A good place to start might be with what is known as a display face. Display faces are those used for headings and tend to be used at a larger size than body type. However, before proceeding, you should familiarise yourself with some of the fundamental concepts of typography along with its associated terminology.

Letterform Anatomy

Start by getting to know the differences between your cap heights and your x-heights and how to tell your serifs from your sans serifs. Whatever you choose to design, it will need to be consistently well styled and proportioned.

Familiarise yourself with the information contained within this chart on the following page:





Types of Typefaces.

“No other design discipline requires so much learning and training as fontography, and by no other aspect can amateurs be so easily distinguished from professionals. To be font literate, a designer has to study the history and the principles of font design.”
(Dmitry Kirsanov)



Classifications

Obviously type comes in thousands of shape variations, and as well as Serifs and Sans Serifs, there are specific classifications for these variations. One such classification system is that created by Maximilien Vox in 1954. Within the current version of the Vox System there are nine groups identified as follows:

1. Classicals

- Humanist
- Garalde
- Transitional

2. Moderns

- 2.1 Didone
- 2.2 Mechanistic
- 2.3 Lineal
 - 2.3.1 Grotesque
 - 2.3.2 Neo-grotesque
 - 2.3.3 Geometric
 - 2.3.4 Humanist

3. Calligraphics

- 3.1 Glyphic
- 3.2 Script
- 3.3 Graphic
- 3.4 Blackletter
- 3.5 Gaelic

Although this system encompasses most of the major categories of type, Cheng (2002) has noted that some of the many typefaces now in existence defy classification.

Humanist, Old Style or Venetian

Humanist typefaces echoed the letter shapes formed by broad nibbed pens.

Transitional

Sharing some of the characteristics of both the humanist and modern faces, transitional faces have medium contrast between the thick and thin parts of the letters.

Modern or Didone

These typefaces exhibit high contrasts between thicknesses. The serifs are referred to as hairlines due to their fineness.

Egyptian

Traditionally, Egyptian or slab serif typefaces were quite heavy with completely squared or bracketed serifs.

Examples for you to consider:

Humanist (Centaur)

Transitional (Baskerville)

Modern (Didot)

Slab Serif (Clarendon)

Technical Options.

"I do not think of type as something that should be readable. It should be beautiful." (Ed Benguiat)



Having decided upon the purpose of your design and having invested a considerable amount of time drawing letters and trying out different variations and combinations, you may wish to attempt to digitise your typeface, effectively making a font.

An experienced designer will probably concur that this part of the process can be as involved—if not more so—than the initial designing of the typeface, particularly if you are striving to perfect the numerous intricacies of the working font.

Let us assume that, as a beginner, you are possibly striving to achieve a basic display face of some kind, at the very least. You might consider making a quirky or decorative typeface. Either way it might be wise to keep in mind the KISS principle of design, meaning ‘keep it simple, stupid’. In other words, don’t try too to achieve too much too soon.

There are several technological choices that you can make when approaching font production and these will largely be determined by your needs and your technical experience.

There is no single definitive design process, and typeface designers approach this somewhat esoteric process in different ways. Some designers work exclusively from screen, without sketching or drawing their designs onto paper, whereas some will scan their drawings and bring these either into font editing software or firstly into Illustrator to be worked on later within a font editor.

If you’re striving to design a display face then you might consider starting out by using what is almost certainly the best beginners’ option for getting quick results: [Fontstruct](#).

Free and simple to use, Fontstruct allows you to use a variety of shaped building blocks in order to form a simple, modular typeface. Once complete, you can then share, download and use the fonts.

FontStruct allows for the design to take place within a web browser, meaning that there’s no requirement to download and install any software. It is incredibly easy to use, and both building and editing are simple. Because of the modular method of construction there are no Bézier curves to manage, which is an easy way to begin.

Once you have completed your design, you can download your font and install it onto your computer. You can also decide whether or not to keep the font private or share it via the FontStruct website. The sharing of fonts within the application means that fellow designers can modify each other’s designs, making this a truly social design experience.

If you want to create typeface that is altogether more sophisticated, then there are more powerful and complex editors such as Fontlab, and the more recently available, Glyphs. Fontlab has been the serious type designer’s tool for a long time, but now has competition from applications such as Glyphs which according to its creator is a “smart and simple approach [that] helps you draw new typefaces, modify existing fonts, and sculpt your letterforms hassle-free.” Unlike some of its competitors, Glyphs has a user friendly bank of tutorials and resources, including an excellent handbook that will help the novice get off to a good start and develop an understanding of many of the refining processes. Unlike FontStruct, Glyphs allows you to draw and edit using Bézier curves, giving you the opportunity to create far more precise shapes.

An exciting new development is Prototypo. Prototypo appears to be a sophisticated browser based tool which is available to download in beta form.

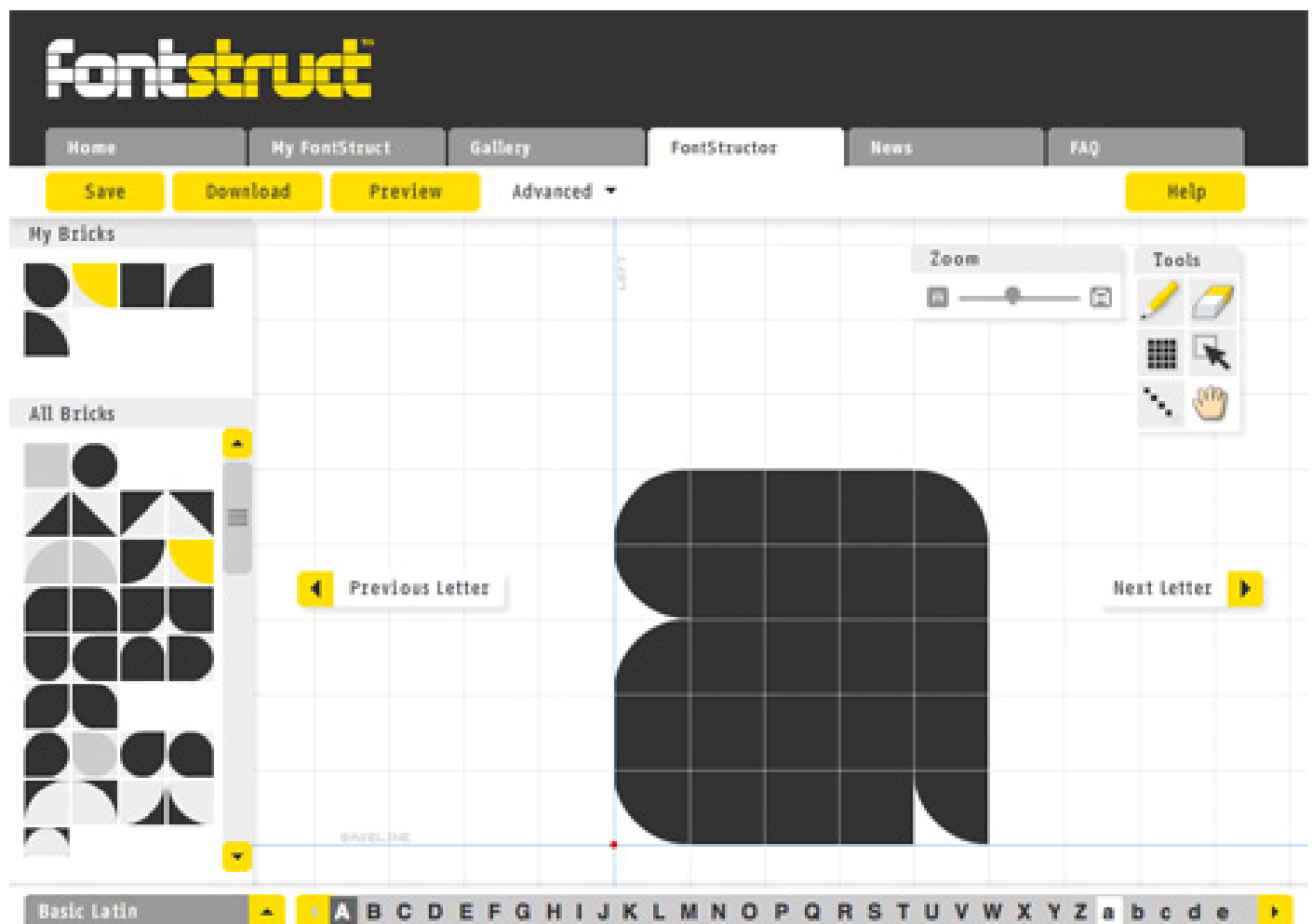
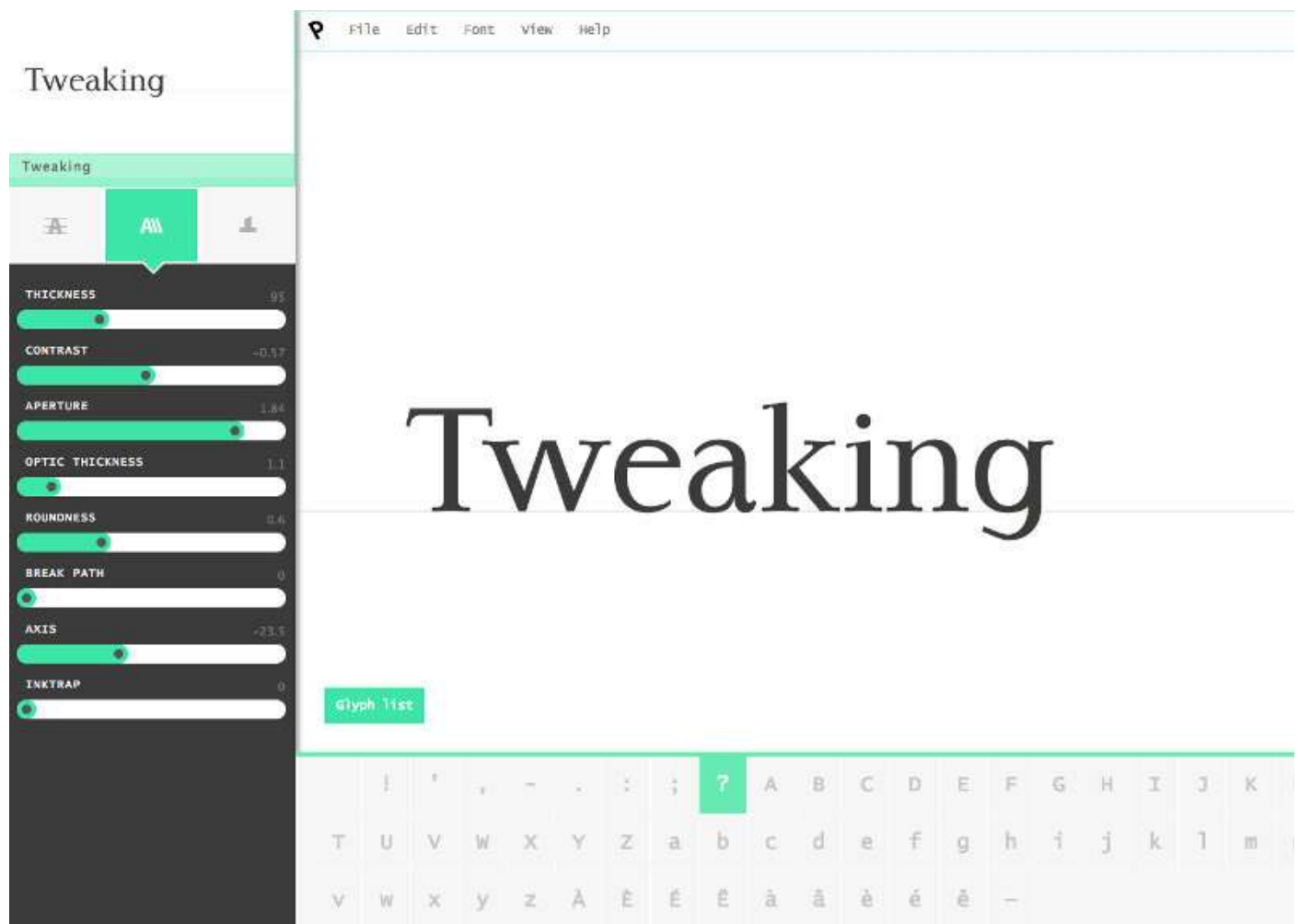
Prototypo

Of course, if you're familiar with Adobe Illustrator (and particularly its pen tool) you can design your letterforms with it. But in order to turn these into working fonts, you'd need to bring them into some font editing software and continue working on them to perfect their shaping and spacing. If you design a typeface using Illustrator you must remember to work in points as this is the true measure of type.

There are several other font editors; some are designed for use on PC and others for Mac such as RoboFont.

If you're looking for a way of digitising your handwriting, then there's MyScriptFont.com which is a free online tool and quite a fun way of generating a font with a personal touch.

Whatever approach you decide to explore, it is good to keep in mind that tools are simply tools and that type design and typography are steeped in centuries of convention. As designer and author David Jury says, *"Rules can be broken but never ignored."*



Understanding Typeface Design.

*“Since typography is a communication method that utilizes a gathering of related subjects and methodologies that includes sociology, linguistics, psychology, aesthetics, and so much more – we aim to educate that there is no single approach within typography that applies to everything.”
(Shelley Gruendler)*



We have touched upon some of the fundamental aspects of typeface design, along with a brief look at some of the type design tools available. To conclude, here are some final points to consider:

Read, read, read.

If you're really keen to develop an understanding of type design, be prepared to learn from those with experience. What has in the past been a fairly enigmatic discipline has become a more open and transparent activity with many people having a go at creating their own designs and sharing their knowledge. Scour books and investigate the many discussion threads in typography forums such as those at typedrawers.com. Specialist libraries are few, but places such as the St. Bride Library in London house an enormous collection of books, type specimens and print artefacts. However, if this is not an option, the reading and references provided below should get you off to a good start.

Draw, or at least doodle, regularly.

You don't have to have great handwriting or be an accomplished calligrapher, however, drawing does enable the immediate conveying of ideas as well as the development of sensitivity as to how shapes and forms interact in relation to each other. Use whatever you have available, be it scraps of paper, napkins or old packaging. Ideally, have a notebook or journal that you can keep with you and utilise as soon as inspiration strikes. A graphics tablet can be a good investment too, as this allows for greater freedom and control than a mouse.

Start with the basics.

Take time to develop sensitivity towards letter shapes. When designing letters, don't just consider the main form of the letter shapes as standalone designs. Instead, consider also the spatial relationship between other letters as well as the spaces inside of the letters (the counters and apertures).

Never steal someone else's design.

Tweaking someone else's design in an attempt to pass it off as your own is unethical and if it breaches copyright law, it's illegal. There are many observant type aficionados who are quick to spot a plagiarised font!

Enjoy it!

Leave the stress to the professionals and simply have fun learning about letters as you play with the possibilities. Good luck.

WILLIAM CASLON

(Baroque 1722 Baroque)

THE FIRST OF A FAMILY of english type founders, was born at Cradley, Worces tershire in 1692, and in 1716 start ed business in London as an engraver of gun locks and bar rels, and as a bookbinder's tool-cutter. Being thus brought into contact with printers, he was induced to fit up a type foundry, largely through the encouragement of printer William Bowyer.

ABCDEFGHIJ KLMNOPQR
STUVWXYZ
abcdefghijkl
klmnopqr
stuvwxyz
1234567
89&?!

ABCDEFGHIJ KLMNOPQR
STUVWXYZ
abcdefghijkl
klmnopqr
stuvwxyz
1234567
89&?!

THE DISTINCTION AND legibility of his type secured him the patron age of the leading printers of the day in England and on the continent. The use of Caslon types, discontinued about the beginning of the 19th century, was revived about 1845 at the suggestion of Sir Henry Cole, and used for printing the Diary of Lady Willoughby (a 17th century story) by the Chiswick Press.

The head line on this page is "Caslon Old Face." He died on the 23rd of January 1766. His son, William Caslon (1720-1778), who had been partner with his father for some years, continued the business. The original printing of the American Declaration of Independence was set with Caslon type, and derivatives of these typefaces continue to be used as computer fonts.

Quoufque tandem
abutere Catilina,
patientia noftra?
quamdiu nos
etiam furor ifte
tuus eludet? quern
ad finem fefe ef-
frenata ja&abit
audacia?

QUOUFQUE TAN-
DEM ABUTERE CA-
TILINA, PATIENTIA
NOFTRA? QUAMDIU
NOS ETIAM FUROR
IFTE TUUS ELUDET?
QUERN AD FINEM
FEFE EFFRENATA
J&ABIT AUDACIA?

Quoufque tandem
abutere Catilina,
patientia nof-
tra? quamdiu nos
etiam furor ifte
tuus eludet? quern
ad finem fefe ef-
frenata ja&abit
audacia?

History History History History
Kolonien Kolonien Kolonien Kolonien
Declaration Declaration Declaration Declaration
Originality Originality Originality Originality
Humanistic Humanistic Humanistic Humanistic
Indepedence Indepedence Indepedence Indepedence
Transparent Transparent Transparent Transparent
Transitional Transitional Transitional Transitional
Experimental Experimental Experimental Experimental

Akzidenz
Grotesk
(1898)

GROTESQUE

IN GERMAN, the word Akzidenz refers to any type of printing other than book and magazine work. Accidenz (sic) Grotesk was released by Berthold in Berlin in 1898, according to their own literature.

Akzidenz-Grotesk is a grotesque typeface originally released by the H. Berthold AG type foundry in 1898 under the name Accidenz-Grotesk. It was the first sans serif typeface to be widely used and influenced many later neo-grotesque typefaces. Max Miedinger at the Haas Foundry used it as a model for the typeface Neue Haas Grotesk, released in 1957 and renamed Helvetica in 1960. Miedinger sought to refine the typeface making it more even and unified. Two other releases from 1957, Adrian Frutiger's Univers and Bauer and Baum's Folio, take inspiration from Akzidenz-Grotesk.

24680
abc defghijklmnopqrstuvwxyz

ABCDEFGHIJ
KLMNOPQR
STUVWXYZ
abcdefghijklmn
opqrstuvwxyz

{ }

The design of Akzidenz-Grotesk was theorized to be derived from Walbaum or Didot, as demonstrated by the similar font metrics when the serifs are removed. However, the font family also included fonts made by other foundries, such as the c. 1880 typeface Royal Grotesk Light from the Berlin foundry Ferdinand Theinhardt Schriftgiesserei, designed by Ferdinand Theinhardt for the scientific publications of the Royal Prussian Academy of Sciences in Berlin.



Glossary.

The following are some of the key terms that pertain to typeface design and typography.



Aperture

An opening of negative space within a letter such as that within the letter 'U' or that within the lower half of the letter 'e'

Ascender

The part of a letter that extends above the x-height. Examples of letters with ascenders include 'b', 'd' and 'h'

Baseline

The invisible line on which type sits

Body Type

Type that forms a main body of text.
(As opposed to a 'display' face)

Counter

A closed space, or partially closed space within a glyph such as that within the letter 'd' or the number 8

Cursive

Letters that join together as in scripted handwriting

Descender

The part of a letter that extends below the baseline. Examples of letters with descenders include 'p' and 'q'

Diacritics

Appearing either above or below a letter, diacritical marks indicate a change in pronunciation. Examples of diacritical marks include the acute (é) and a circumflex (ê)

Display Type

Type that is generally of a large size. Often this can be easily read at a distance or used in headings.

Fount (archaic UK) Font (US)

The physical means by which a typeface is applied or rendered. This normally takes the form of either a digital or metal set of letters and glyphs. Example: Gill Sans Light 10pt

Glyph

A symbol or character

Hinting

Part of the type design process whereby type is lined up against a pixel grid

Italic

A sloping type based on cursive handwriting.

Kerning

The adjustment of space between two letters

Legibility

"A measure of a type's ability to be read under normal reading conditions." (White, 2004)

Ligatures

A glyph that is formed by joining two or more glyphs

Majuscules

Capital or uppercase letters

Minuscules

Lowercase letters

Multiple Mastering

A technological facility that allows for the creation of a variety of font weights from one master

Opentype

A font file format that works across platforms. It generally has the suffix .otf

Pica (Py)

Consisting of 12 points, a pica is typographic measure expressed.

Point (Pt)

A single unit of a measure of type. There a 12 points to a pica.

Readability

“A measure of the type’s ability to attract and hold a reader’s interest. Increasing readability often causes a reduction in legibility...” (White, 2004)

Serif

A small stroke or addition to the end of letter or glyph stroke. These come in a variety of shapes and styles such as hairline, slab, bracketed and wedge.

Sans Serif

Type that lacks the addition of serifs.

Sidebearings

This term refers to the spaces either side of a glyph.

Swash

An added flourish or embellishment such as a lengthened tail or exaggerated, decorative serif.

True Type Font

or .ttf is a forerunner of the Opentype format.

Tracking

The adjustment of space between multiple letters or tracts of text.

Typeface

The design of a set of an alphabet, numbers, punctuation, etc.

Tail

A descending, often extended stroke the generally runs from left to right such as that of the letter ‘Q’

Terminal

The end of a stroke without a serif. Example of a stylised version is the ball terminal

Weight

Refers to the relative density and stroke widths of a given typeface, such as thin, regular, bold, extra bold, etc.

X-Height

The height of a given lowercase 'x'

Alan Cumming, Polymath.
Anything goes when
you get into an elevator
with the star of *The
Good Wife* and *Cabaret*



BY DREW GRANT
PHOTOS BY CHRIS CRISMAN

Shot on location at
Geddy Rose
Recording Studios.
Hair and makeup
by Michael Murren



Further Reading.

BOOKS

Bringhurst, R., The Elements of Typographic Style

A seminal typographic text, Bringhurst's book is a thorough and expansive wealth of typographic information and is ideal for both the beginner as well as the advanced practitioner.

Cheng, K., Designing Type

Cheng's book is an incredibly useful resource. This is not a 'how to book' but is a great reference for understanding the structure and systems of type. Highly illustrated throughout.

Earls, D., Design Typefaces

Harvey, M., Lettering Design

Heller, S., Typography Sketchbooks

A compendium of typographic designers' sketchbooks.

White, A., Thinking in Type: The Practical Philosophy of Typography

Jury, D., About Face: Reviving the Rules of Typography

MAGAZINES

Font is a free - online magazine produced by Fontshop

Baseline Magazine

WEB

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Briem, G., Notes on Type Design

De Gregorio, J.P., Vector Drawing

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Shaw, P., Ten Simple Rules for Researching Letterforms

Stössinger, N., Sketching Out of My Comfort Zone: A Type Design Experiment

On drawing Type by TypeCooker.com

- Type Anatomy

FORUMS

<http://typedrawers.com/> – "Typedrawers is a forum where fellow typophiles can gather to discuss typography."

GAMES

Kerntype - <http://type.method.ac/>

Shapetype - <http://shape.method.ac/>



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